Question	Answer	Marks	Partial Marks
1(a)(i)	600	**2**	**M1** for $\frac{1250}{12+9+4}\times k$ where k = 1, 4, 9, 12 oe
1(a)(ii)	80	**2**	**M1** for 1250 × 64 [÷ 1000]
1(a)(iii)	60	**2**	**M1** for $x\times\left(1-\frac{10}{100}\right)=54$ oe
1(a)(iv)	1000	**2**	**M1** for 1250 – (1250 ÷ 5) oe or **B1** for 250
1(b)(i)	3.52	**2**	**M1** for [10 –] 12 × 0.54 or **B1** for 6.48
1(b)(ii)	0.08	**3**	**B2** for 0.077[4...] or **M1** for 0.51 ÷ 0.826 If 0 or 1 scored award instead **SC2** for 0.93 final answer OR If 0 scored **SC1** for 0.06 as answer
2(a)	[sin =] $\frac{145}{\frac{1}{2}\times 6.4\times 5.7\times 15}$	**M2**	**M1** for 145 = $\frac{1}{2}\times 6.4\times 5.7\times \sin x\times 15$ oe or for $\frac{1}{2}\times 6.4\times h\times 15=145$ and $\sin x=\frac{h}{5.7}$
	32.0[0]	**A1**	If M0, **SC1** for 145 = $0.5\times 6.4\times 5.7\times sin32\times 15$ oe
2(b)	3.4[0] or 3.402 to 3.403 nfww	**3**	**M2** for $\sqrt{6.4^2+5.7^2-2\times 6.4\times 5.7\times\cos(32)}$ OR **M1** for $6.4^2+5.7^2-2\times 6.4\times 5.7\times\cos(32)$ **A1** for 11.6 or 11.57 to 11.58
2(c)	3.02 or 3.020 to 3.021	**3**	**M2** for $\sin(32)=\frac{x}{5.7}$ $\sqrt{80^2+50^2-2\times 80\times 50\times\cos 75}$ or **M1** for recognition that the line from E is perpendicular to AB e.g. right angle seen or $\frac{1}{2}\times 6.4\times h$

Question	Answer	Marks	Partial Marks
2(d)	10.8 or 10.9 or 10.84 to 10.85...	**4**	**M3** for [sin =] $\frac{their\,(c)}{\sqrt{15^2+5.7^2}}$ or $\left[\tan =\right]\frac{their\,(c)}{\sqrt{\left(5.7\times cos32\right)^2+15^2}}$ or **M2** for $15^2+5.7^2$ or $\left(5.7\times\cos 32\right)^2+15^2$ oe or **M1** for recognition of correct angle
2(e)	136 or 136.0...	**3**	**M2** for $938\times145\times\frac{1000}{1000000}$ oe or **M1** for figs 136 or 13601
3(a)(i)	55.87	**4**	**M1** for midpoints soi **M1** for use of $\sum fm$ where m is in the correct interval including boundaries **M1** (dep on 2nd M1) for $\sum fm \div 1000$
3(a)(ii)	$\frac{177}{500}$ cao	**2**	**M1** for $\frac{154+200}{1000}$ oe
3(b)(i)	25000	**1**	
3(b)(ii)	2.473×10^4	**1**	
3(c)(i)	166 650 or 165816 nfww	**3**	**M2** for (500 + 5) × '320 to 340' or '500 to 510' × (320 + 10) or **M1** for 500 − 5 or 500 + 5 or 320 -10 or 320 $+10$ Alternative method **M2** for 504 × '320 to 340' or '500 to 510' × 329 or **M1** for 504 or 329
3(c)(ii)	285 or 286 nfww	**2**	**M1** for 800 -10
4(a)(i)	96	**2**	**M1** for $\frac{1}{2}\times24\times8$
4(a)(ii)	18.4 or 18.43...	**2**	**M1** for $\tan\left[x\right]=\frac{8}{24}$ oe